

Official Statistics Data Quality Audit of Local Authority Housing Affordability Indicators in England and Wales, 2002-2025

A public evidence analytics portfolio paper

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Tools: Python, DuckDB SQL, pandas, matplotlib

Abstract. This paper presents a compact official-statistics-style audit of local-authority housing affordability indicators. The analysis uses the Office for National Statistics house price to residence-based earnings ratio dataset, focusing on sheet 5c: the ratio of median house price to median gross annual residence-based earnings by local authority district for England and Wales from 2002 to 2025. The audit checks structural validity, missingness, geography-code consistency, outliers and comparability risks before producing an analysis-ready long-format dataset. The workflow demonstrates practical skills required in public-sector analytics roles: Python data preparation, SQL validation, reproducible outputs, statistical communication and clear limitation handling.

Keywords: official statistics; data quality; housing affordability; local authorities; Python; SQL; public-sector analytics

1. Why this project matters

Housing affordability is a live policy issue because it links household incomes, local housing markets and geographic inequality. A headline affordability ranking can be misleading if users do not first understand coverage gaps, missing values, outliers and boundary-related comparability issues.

The objective here is not to produce a causal housing model. It is to show the analytical discipline expected in official-statistics and public-sector data roles: check the data before analysis, preserve uncertainty, flag records that require review and communicate findings without overstating the evidence.

The project is designed as a work sample for roles in organisations such as ONS/UK Statistics Authority, Government Social Research, NatCen, NHS/UKHSA analytics and health or housing policy research teams.

2. Data source and audit design

Data source. The audit uses the ONS house price to residence-based earnings ratio dataset. ONS describes this dataset as affordability ratios calculated by dividing house prices by gross annual residence-based earnings, based on median and lower-quartile house prices and earnings in England and Wales. The release used for this portfolio was published on 26 March 2026 and contains 2025 values.

Selected sheet. Sheet 5c was selected because it gives the ratio of median house price to median gross annual residence-based earnings by local authority district from 2002 to 2025. This gives one record per local authority in wide form, with year columns that can be reshaped for analysis.

Audit checks. The workflow checks row and column structure, duplicate rows, local authority code pattern validity, missing values by year and geography, year-specific statistical outliers using the interquartile range rule, regional summaries and changes between 2002 and 2025.

3. Audit summary

Metric	Value
Total local authorities in source sheet	318
Total year-level records after reshaping	7632
Years covered	2002 to 2025
Missing house price-to-earnings ratio records	61
Statistical outlier records flagged	110
Local authorities with at least one missing year	23
Local authorities with valid code pattern	318
Invalid local authority code patterns	0
Duplicate local authority codes	0
Areas with both 2002 and 2025 values	295
Average affordability ratio in 2002	5.11
Average affordability ratio in 2025	7.98
Average affordability-ratio increase, 2002 to 2025	2.87

Table 1. Final audit summary generated from the notebook workflow.

The source sheet contained 318 local authorities. After reshaping the data from wide to long format, the analysis dataset contained 7,632 year-level records. All local authority codes followed the expected English/Welsh area-code pattern and no duplicate local authority codes were found. The principal quality issues were missing ratio values and statistical outliers.

4. Missingness and coverage

The audit identified 61 missing year-level ratio values. Twenty-three local authorities had at least one missing year. Missingness was concentrated at the start of the time series: 2002 alone had 22 missing values. The largest area-level gaps were Isles of Scilly and City of London, which should be treated cautiously in long-period comparisons.

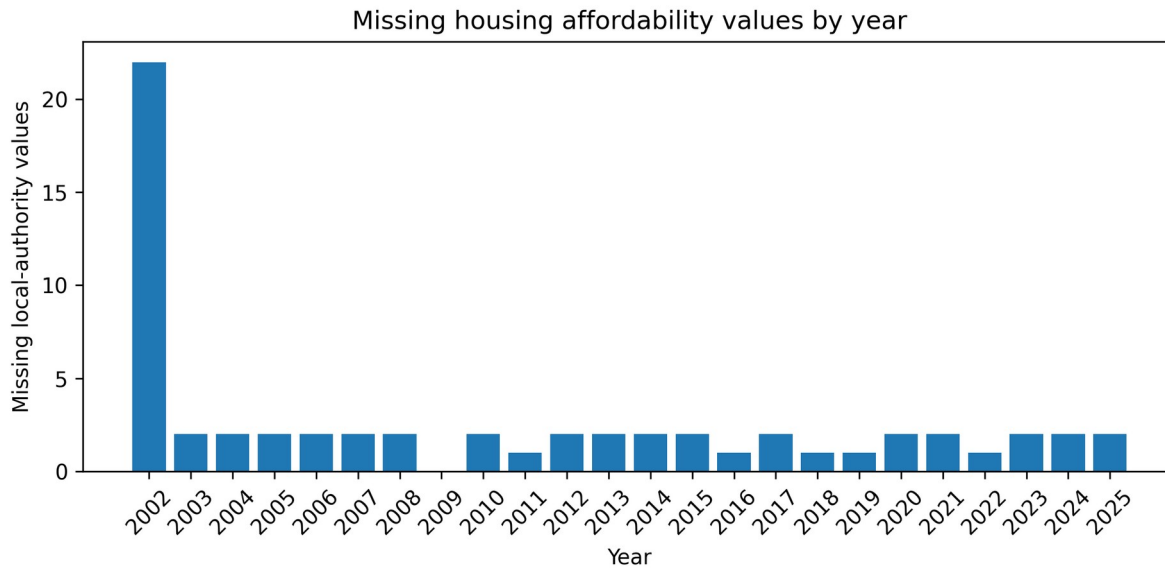


Figure 1. Missing housing affordability values by year.

5. Outlier audit

The notebook flagged 110 year-level outlier records using a year-specific interquartile range rule. These were not treated automatically as errors. In a housing affordability dataset, recurring outliers may represent valid local market extremes rather than bad data. The quality approach is therefore to flag and preserve these records, not delete them.

Region	Code	Local authority	Outlier years	First year	Last year	Max ratio	Average outlier ratio
London	E09000033	Westminster	23	2002	2025	24.79	17.57
London	E09000020	Kensington and Chelsea	22	2002	2025	34.77	21.99
London	E09000013	Hammersmith and Fulham	20	2005	2025	20.04	15.77
London	E09000007	Camden	18	2008	2025	19.65	16.42
London	E09000012	Hackney	5	2014	2020	17.29	15.44
London	E09000019	Islington	4	2013	2016	16.51	14.73
London	E09000027	Richmond upon Thames	3	2014	2022	16.60	14.56
North West	E07000117	Burnley	3	2004	2006	3.00	2.41
South West	E06000053	Isles of Scilly	2	2009	2019	25.28	21.12
East of England	E07000098	Hertsmere	2	2014	2022	16.58	14.84

Table 2. Local authorities most frequently flagged as statistical outliers.

6. Descriptive findings for 2025

In 2025, regional median ratios were highest in London, followed by the South East, East of England and South West. The lowest regional median ratio was in the North East. This regional gradient is consistent with the concentration of high affordability pressure in southern England, particularly London and its surrounding markets.

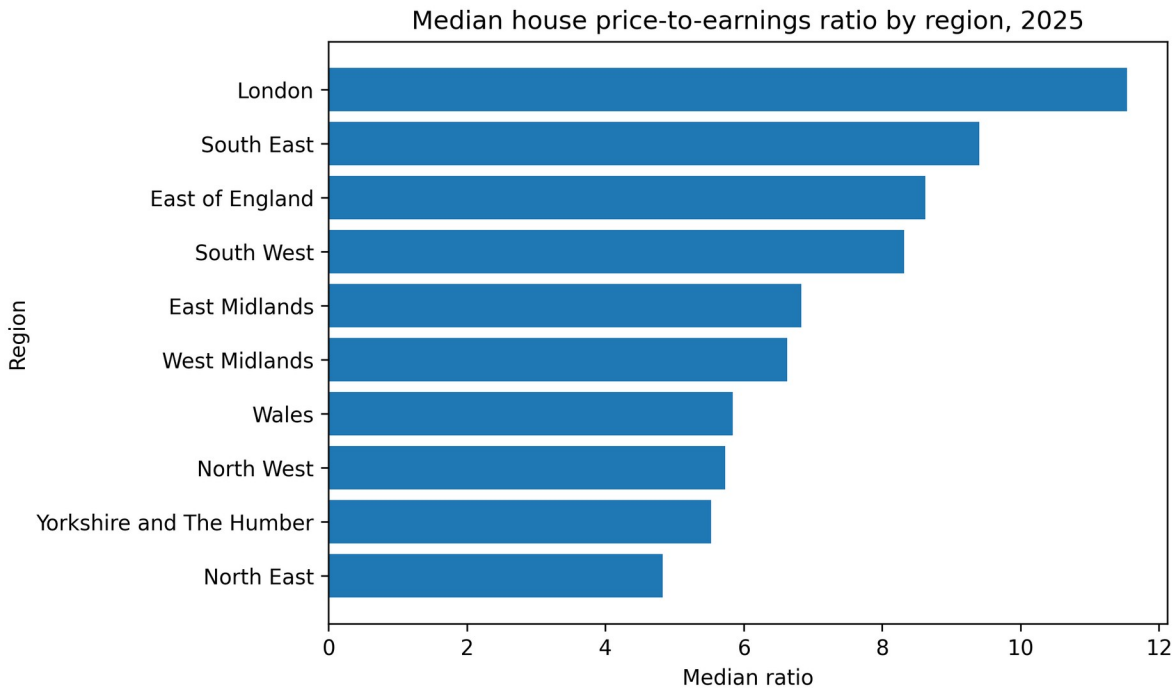


Figure 2. Regional median house price-to-earnings ratios in 2025.

Region	Local authorities	Mean ratio	Median ratio	Min ratio	Max ratio
London	32	11.95	11.54	9.59	21.87
South East	64	9.43	9.40	6.82	13.51
East of England	45	8.73	8.62	6.32	12.39
South West	26	8.31	8.32	6.39	11.03
East Midlands	35	6.73	6.83	5.36	7.96
West Midlands	30	6.95	6.63	4.73	9.17
Wales	22	5.79	5.84	3.98	7.46
North West	35	5.76	5.73	3.86	8.67
Yorkshire and The Humber	15	5.70	5.53	4.46	7.53
North East	12	4.92	4.83	4.28	5.89

Table 3. Regional summary of local authority affordability ratios, 2025.

7. Local authority rankings in 2025

The least affordable local authorities in 2025 were dominated by London boroughs. Kensington and Chelsea had the highest ratio in the audit output, followed by Westminster, Hammersmith and Fulham and Camden. The most affordable local authorities included Hyndburn, Burnley, Merthyr Tydfil and Blaenau Gwent.

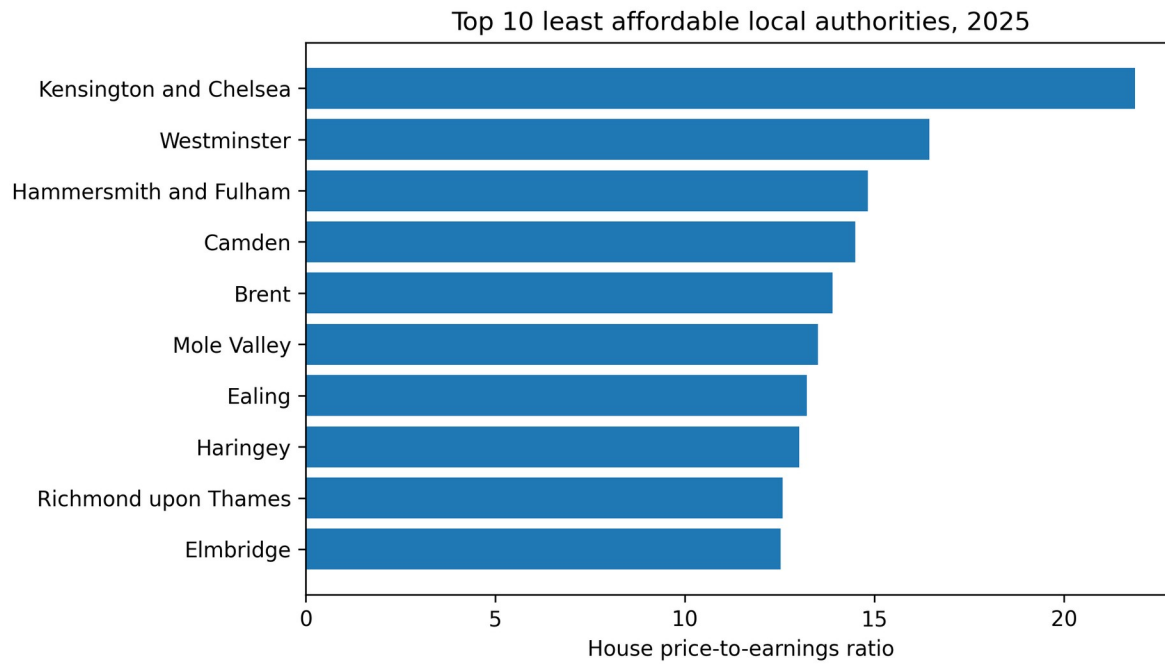


Figure 3. Top 10 least affordable local authorities in 2025.

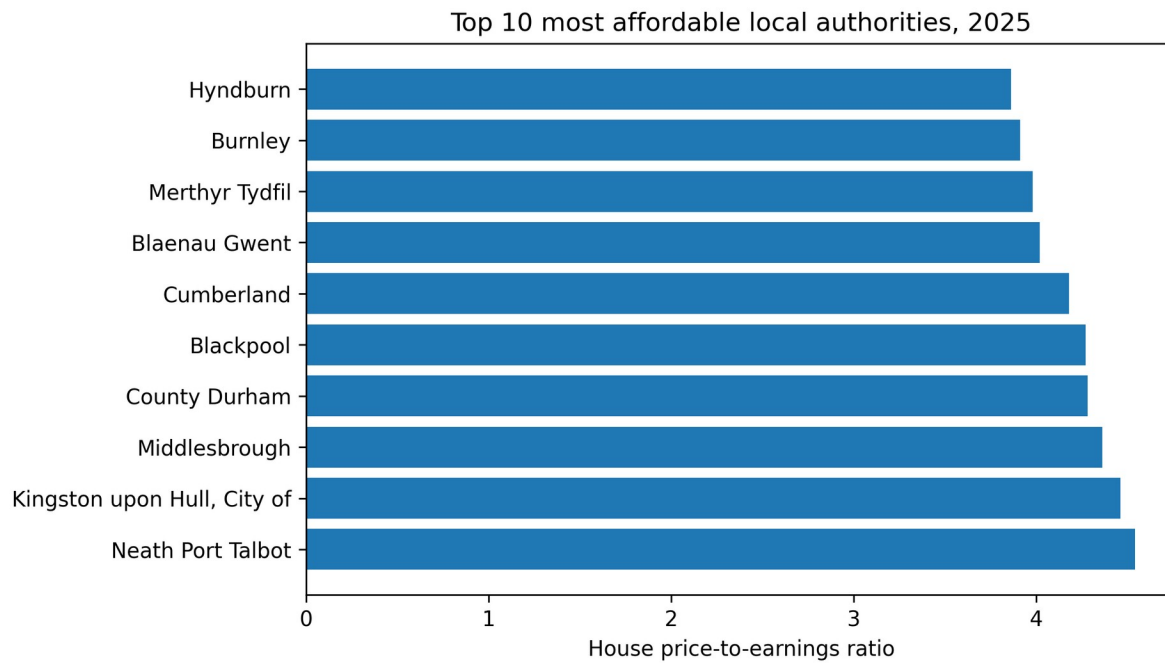


Figure 4. Top 10 most affordable local authorities in 2025.

Region	Code	Local authority	Ratio
London	E09000020	Kensington and Chelsea	21.87
London	E09000033	Westminster	16.44
London	E09000013	Hammersmith and Fulham	14.83
London	E09000007	Camden	14.49
London	E09000005	Brent	13.89
South East	E07000210	Mole Valley	13.51
London	E09000009	Ealing	13.21
London	E09000014	Haringey	13.02
London	E09000027	Richmond upon Thames	12.58
South East	E07000207	Elmbridge	12.52

Table 4. Top 10 least affordable local authorities, 2025.

Region	Code	Local authority	Ratio
North West	E07000120	Hyndburn	3.86
North West	E07000117	Burnley	3.91
Wales	W06000024	Merthyr Tydfil	3.98
Wales	W06000019	Blaenau Gwent	4.02
North West	E06000063	Cumberland	4.18
North West	E06000009	Blackpool	4.27
North East	E06000047	County Durham	4.28
North East	E06000002	Middlesbrough	4.36
Yorkshire and The Humber	E06000010	Kingston upon Hull, City of	4.46
Wales	W06000012	Neath Port Talbot	4.54

Table 5. Top 10 most affordable local authorities, 2025.

8. Change between 2002 and 2025

Across local authorities with both 2002 and 2025 values, the average house price-to-earnings ratio increased from 5.11 in 2002 to 7.98 in 2025, an average absolute increase of 2.87 ratio points. The largest increases were again concentrated in London and parts of the South East and East of England.

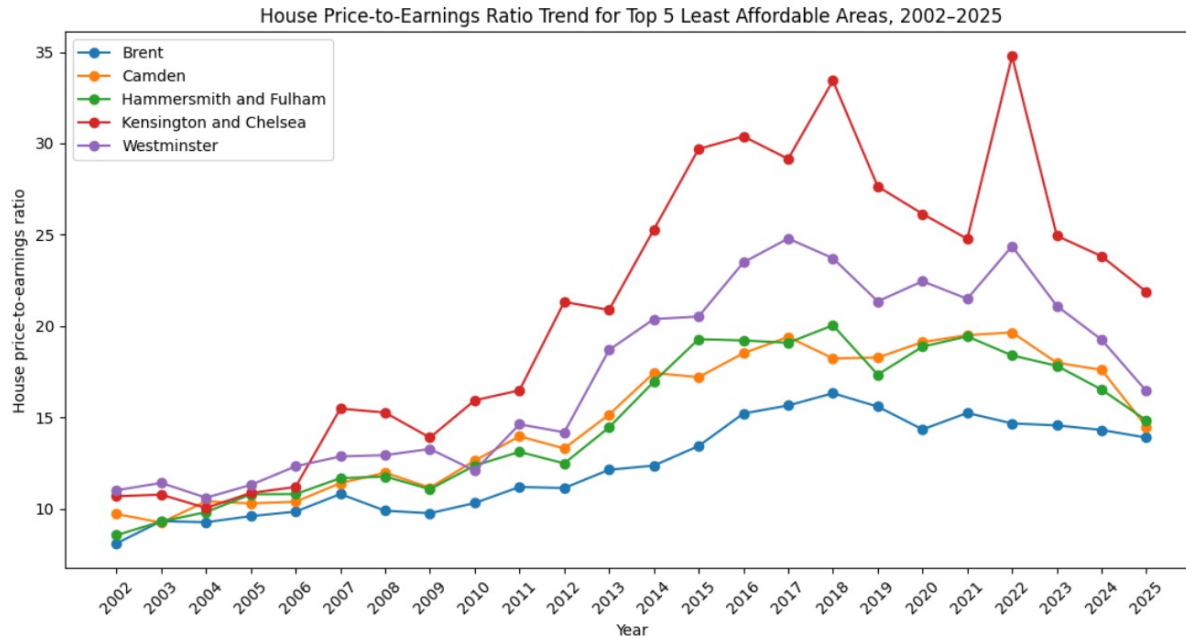


Figure 5. Trend for the five least affordable local authorities in 2025, using the notebook chart output.

Region	Local authority	Ratio 2002	Ratio 2025	Absolute change	Percent change
London	Kensington and Chelsea	10.68	21.87	11.19	104.78
London	Hammersmith and Fulham	8.55	14.83	6.28	73.45
London	Waltham Forest	5.69	11.86	6.17	108.44
London	Ealing	7.08	13.21	6.13	86.58
London	Haringey	7.03	13.02	5.99	85.21
London	Brent	8.10	13.89	5.79	71.48
South East	Mole Valley	7.79	13.51	5.72	73.43
East of England	Hertsmere	6.64	12.27	5.63	84.79
South East	Chichester	6.16	11.77	5.61	91.07
London	Hillingdon	6.47	12.07	5.60	86.55

Table 6. Largest absolute increases in local authority affordability ratios, 2002 to 2025.

Region	Local authority	Ratio 2002	Ratio 2025	Absolute change	Percent change
North West	Blackpool	3.51	4.27	0.76	21.65
South West	East Devon	7.54	8.36	0.82	10.88
South East	Gosport	5.67	6.82	1.15	20.28
South West	Forest of Dean	5.33	6.50	1.17	21.95
East of England	South Cambridgeshire	7.54	8.75	1.21	16.05
North West	Wyre	3.91	5.13	1.22	31.20
East Midlands	Rutland	6.72	7.94	1.22	18.15
West Midlands	Solihull	5.78	7.04	1.26	21.80
South West	Torbay	6.16	7.46	1.30	21.10
North East	Sunderland	3.32	4.64	1.32	39.76

Table 7. Smallest absolute increases in local authority affordability ratios, 2002 to 2025.

9. Quality flags and reproducible workflow

The audit assigns each year-level record one of three flags: OK, Missing ratio value or Statistical outlier. This approach preserves the original values while making downstream analysis safer. It avoids two common errors: silently dropping inconvenient records and treating every statistical outlier as a data error.

The long-format quality-flagged dataset can be used for further analysis, visualisation or dashboarding. The SQL script provides employer-visible evidence of record counting, missing-value aggregation, 2025 ranking, regional summaries and extraction of records requiring review.

The workflow is intentionally simple and transparent. It uses Python for reading, cleaning and reshaping; DuckDB SQL for validation and summaries; and matplotlib for charts. This is appropriate for a public-sector analyst work sample where auditability and clear communication matter more than model complexity.

10. Limitations

- This is a data quality and descriptive analysis project, not a causal study of housing markets.
- The outlier method is statistical. It does not establish whether an extreme value is erroneous; it only marks it for review.
- Missing values were not imputed. They were preserved and flagged to avoid artificial precision.
- Boundary changes and local government reorganisation may affect comparability across the full 2002-2025 period.
- The ranking uses the median house price-to-residence-based earnings ratio only. It does not include mortgage rates, rents, deposits, household wealth, household composition or first-time buyer constraints.

References

- [1] Office for National Statistics. House price to residence-based earnings ratio dataset. Released 26 March 2026. Dataset description: affordability ratios calculated by dividing house prices by gross annual residence-based earnings, based on median and lower-quartile house prices and earnings in England and Wales.
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